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Photoacoustic Communication from the Air to Underwater Based on Low-Cost Passive Relays

SREEHARI M-VAS21EC097

Seminar Coordinators :

Resmi K.Rajan, Asst. Professor, DoECE

Divya Unni, Asst. Professor, DoECE

Jemy Jose Kakkassery, Asst. Professor, DoECE

**Vidya Academy of Science & Technology
Department of Electronics and Communication Engineering**

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- With the development of ocean research, communications from the air to underwater (e.g., aircraft to submarine) are especially important. However, the existing communication methods have their own limitations and wireless communication between underwater and the air has remained an unsolved problem.
- In this paper, we see a photoacoustic communication system based on low cost passive relays, which is capable of not only minimizing the energy loss both in water and in the air-water interface, but also relieving the difficulty of target seeking.
- Thus, we can get a six order of magnitude reduction of laser energy (25J vs. 27 μ J) using passive relays.
- It can work decently under complex condition like pollution and waving.



KEY TERMS

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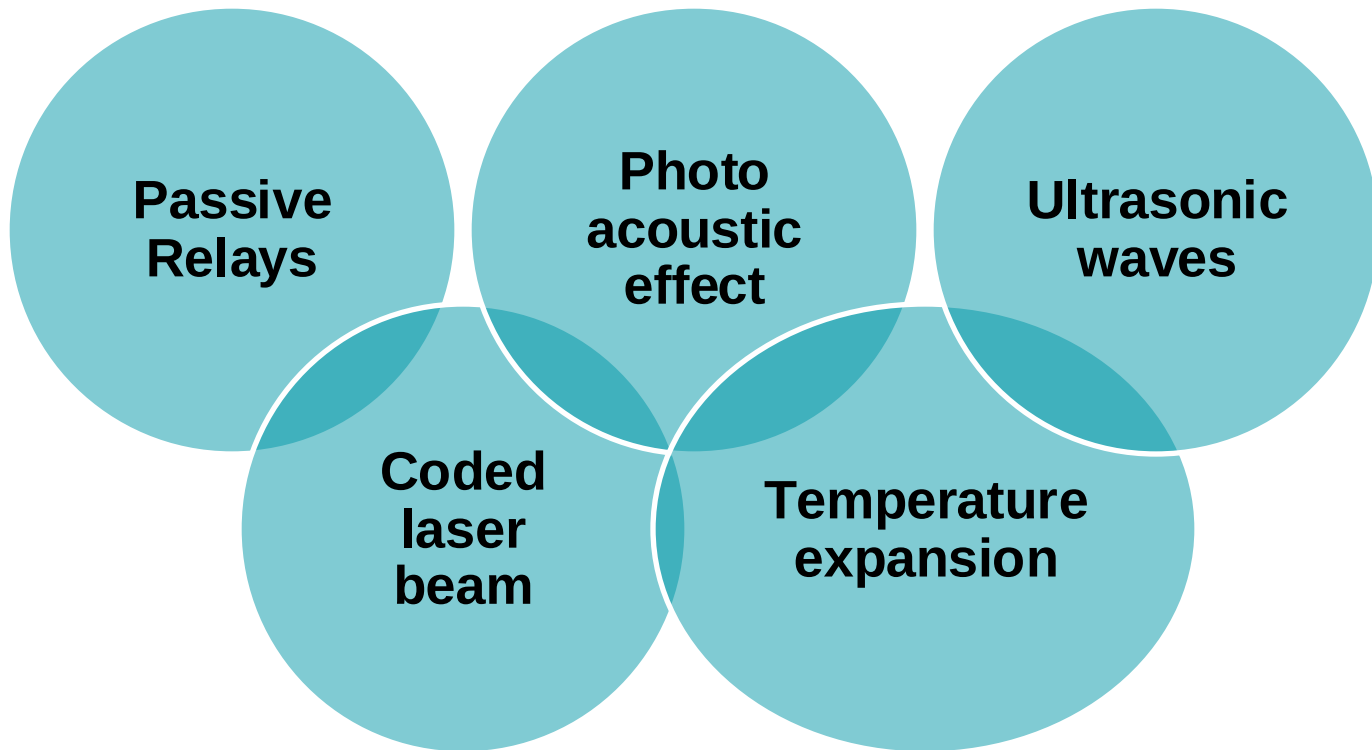
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- Wireless communication between underwater and air has been a longstanding challenge, crucial for marine technology and underwater exploration.
- In 2018, an MIT group developed one-way communication from underwater to air using translational acoustic-radio frequency communication technology.
- Air-to-underwater communication still faces significant limitations.
- Direct communication is challenging due to issues such as energy loss and low stability of light communication at the air-water interface, high attenuation of electromagnetic waves in water, and substantial energy loss of ultrasonic waves due to the large acoustic impedance difference at the interface.
- Active relays, which convert air signals to underwater signals. They are installed in offshore base stations, buoys, or ships.



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- But they have high cost and maintenance requirements, incomplete ocean coverage leading to potential communication failures. And the large volume raises security concern. As a result, passive relays can be used.
- Therefore, there is an urgent need for passive relays to improve air-to-underwater communication efficiency.

BLOCK DIAGRAM

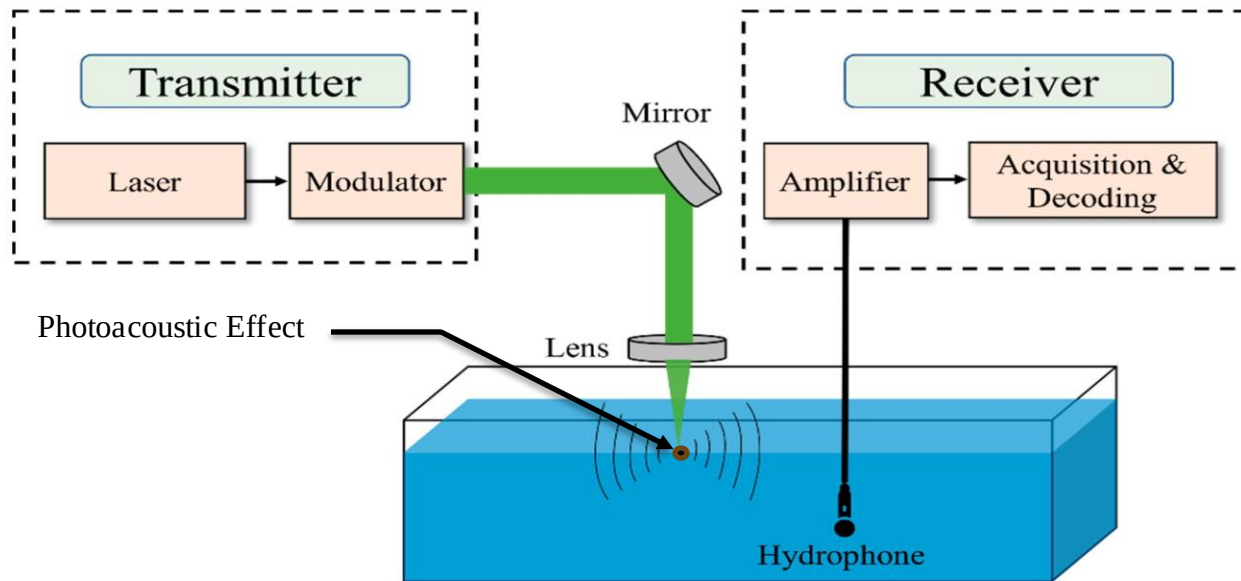


Figure 1. Block diagram of photoacoustic air-underwater wireless communication

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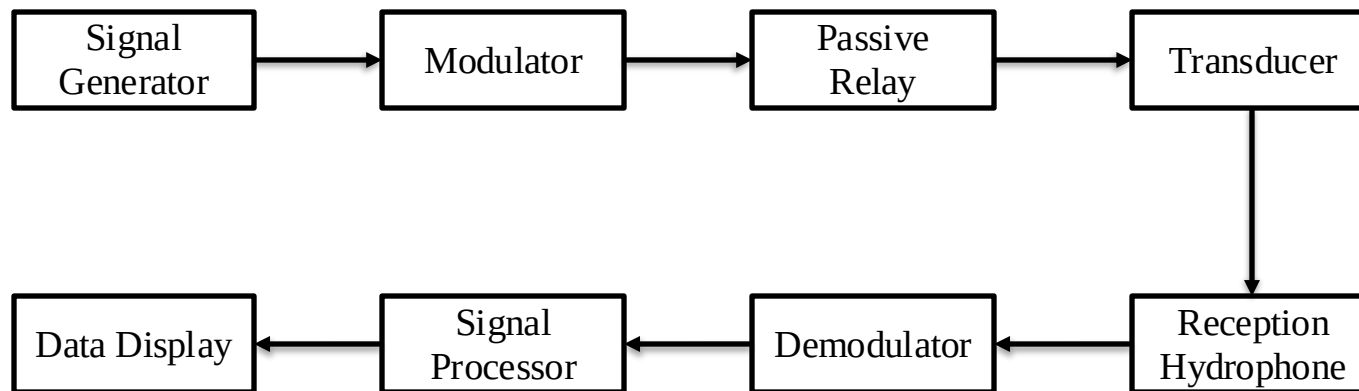


Figure 2. Block diagram of photoacoustic air-underwater wireless communication



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PHOTOACOUSTIC EFFECT

- The photoacoustic effect, also called as optoacoustic effect, in the usual sense is a more efficient way to convert light energy into ultrasonic energy at light absorbers by thermal expansion.
- When the absorber is illuminated with a pulsed laser, a small temperature rise can produced to generate a photoacoustic wave under the conditions of thermal and stress confinement.
- These are widely used in biomedical fields and industrial applications.
- Photoacoustic wave have less attenuation than electromagnetic waves.
- Photoacoustic waves travel more efficiently in water than electromagnetic waves.



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PASSIVE RELAYS

- Passive relays are devices used in communication systems to extend the range or improve the coverage of a signal without requiring an external power source.
- Unlike active relays, which amplify and retransmit signals using powered electronic components, passive relays simply reflect, redirect, or otherwise manipulate the existing signal to achieve better propagation.
- They are used between the transmitter and the receiver to extend the range, redirect signal, enhance to a specific frequency.
- They are not a part of the transmitter or the receiver, but plays a crucial role in managing the propagation of acoustic waves in complex environments.



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- Passive relays are often used in environments where it is difficult or impractical to provide power.
- Low-cost and high-efficiency relays proposed to improve communication efficiency.
- They can be used for significant reduction in laser energy required (from 25J to 25 μ J) with low-cost passive relays.
- Passive relays cost less than US \$1 (₹ 83.94)
- Applications for passive relays include enhancing wireless communication in urban areas, improving radar coverage, and extending the reach of optical communication systems.



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- In the photoacoustic air-to-underwater communication system involves a sender in the air, such as an aircraft, emitting a pulsed laser beam carrying coded information.
- This laser beam then strikes a signal relay, which is a light-absorbing material released by a receiver, such as a submarine.
- The signal relay generates pulsed ultrasonic signals, known as photoacoustic waves, due to the photoacoustic effect.
- These waves are detected by the receiver underwater, which decodes the information.

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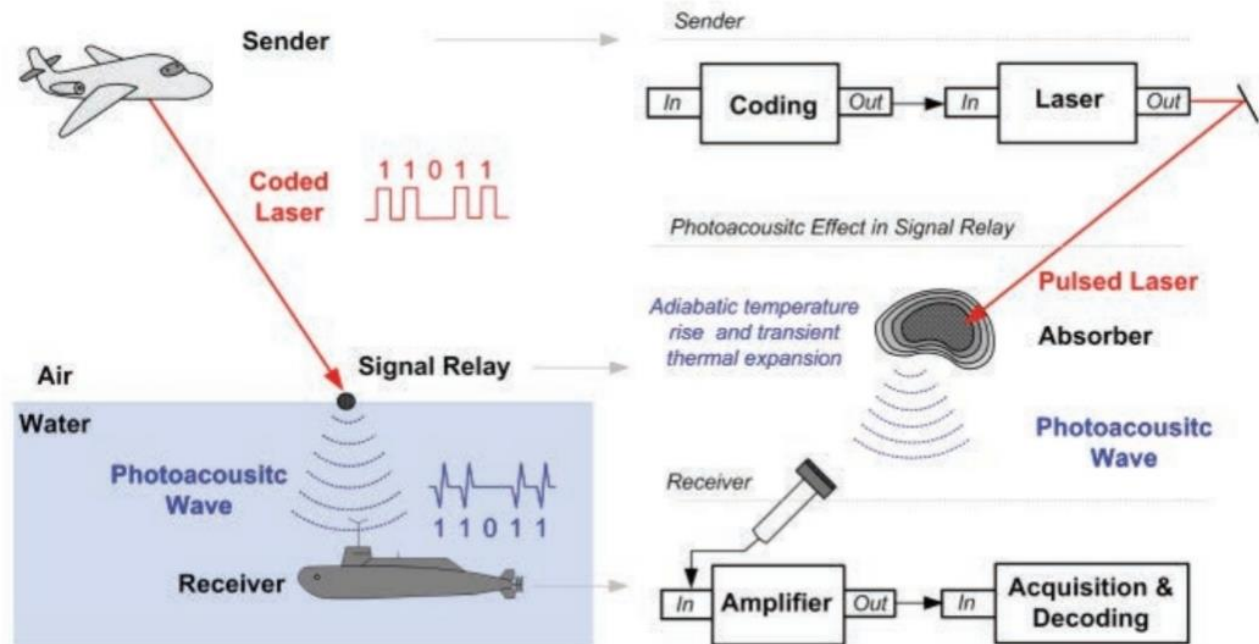


Figure 3. Block diagram of photoacoustic air-underwater wireless communication based on low-cost passive relays



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- The photoacoustic effect occurs when a pulse laser illuminates the absorber, causing a small, transient temperature rise.
- This leads the heated absorber to thermally expand and contract, generating photoacoustic waves. This entire process is estimated by an experimental set up.
- In the experimental setup, a custom-made laser (Beijing Laize, China) emits a 660 nm wavelength laser beam with a 14 ns pulse width at a repetition frequency of 1 kHz.
- The function generator and the laser acts as the information sender.
- The laser has a single pulse energy of 27 mJ.
- The ultrasonic signal receiver is composed of an ultrasonic detector (5 MHz), a home-made amplifier (40 dB magnification), an oscilloscope (Tektronix MD)3054), and a computer for decoding signals.



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- When the laser beam reaches the signal relay, ultrasonic waves are generated by the photoacoustic effect.
- These photoacoustic waves travel underwater and are received by the ultrasonic detector. The signal is then amplified, acquired by the oscilloscope, and decoded by the computer.
- The system generates photoacoustic signals with voltages ranging from approximately 100 mV to 1 V, achieving a six-order-of-magnitude reduction in laser energy (27 mJ vs. 25 J) using low-cost passive relays, which cost less than US\$1.00.
- This approach offers several advantages, including the use of a low-power light source and low-cost passive relays, enhancing efficiency and reducing costs.



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- It provides high communication efficiency with significantly reduced laser energy requirements.
- However, high laser power may cause photobleaching of absorbers, shortening communication duration, and further increases in light energy can lead to optical breakdown of water, which is another topic.



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- Photoacoustic communication, by converting light into ultrasound, provides a reliable alternative to pure optical methods in challenging water environments.
- Unlike light, which is prone to direction shifts and absorption in wavy or muddy water, photoacoustic waves maintain stable transmission and are less affected by such conditions.
- Experiment demonstrate that while light signals struggle in wave conditions and polluted water, photoacoustic waves exhibit minimal energy loss and remain effective.
- A proof-of-concept experiment with binary code transmission further underscores the system's ability to decode signals into readable information, proving its potential for more complex applications.
- Overall, photoacoustic communication is validated as a robust solution for air-to-underwater communication where traditional optical methods fail.

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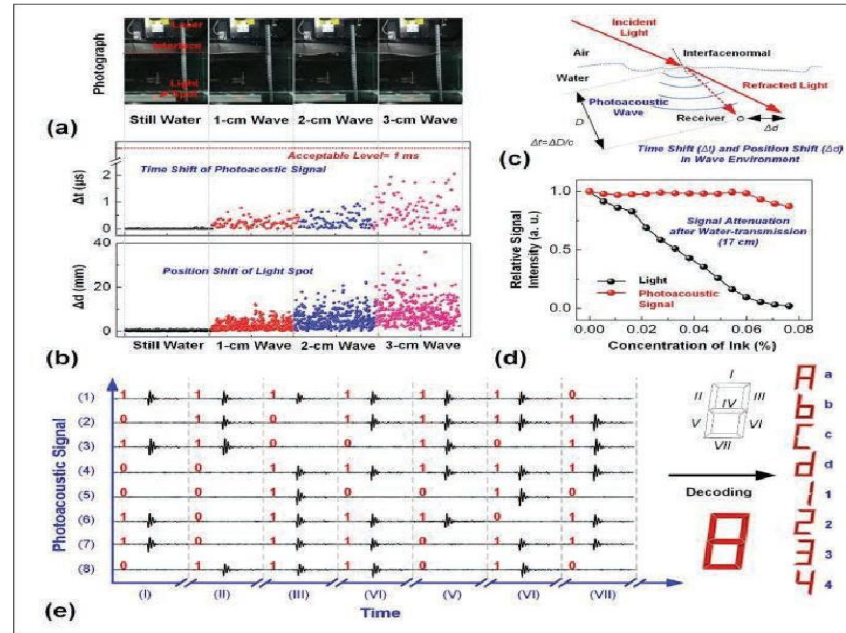


Figure 4. (a) Photographs of Communication path in 1cm, 2cm, 3cm waves (b) Measured time shift of photoacoustic wave (Δt) and position shift of light spot (Δd) in different magnitude of waves (c) signal propagation path (d) relative intensity of light and photoacoustic waves (e) Raw time-domain signal decoding

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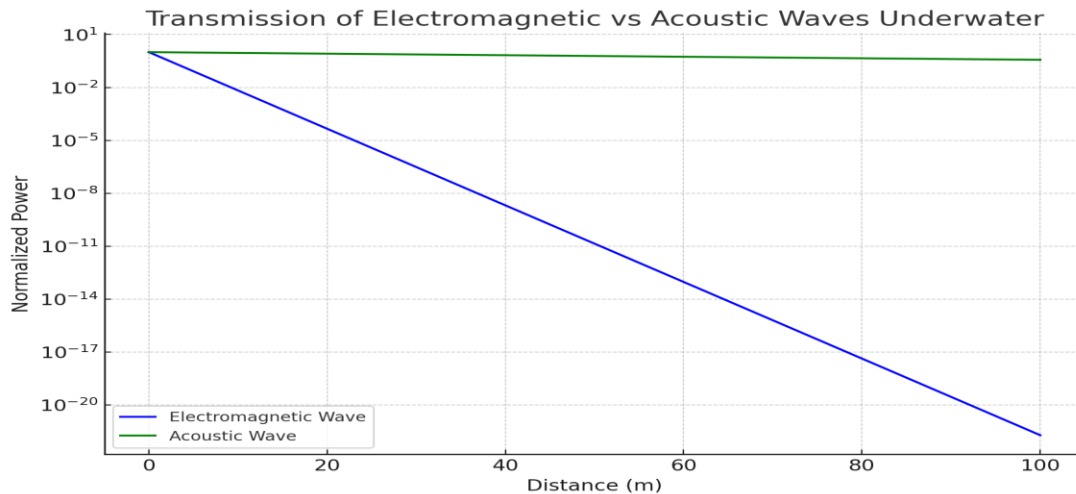


Figure 5. Comparison of Underwater Acoustic wave and Electromagnetic wave attenuation



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1.Enhanced Signal Clarity

Future developments could improve signal clarity by optimizing relay materials and designs to minimize distortion and attenuation during the air-to-water transition.

2.Advanced Relay Materials

Innovations in materials science may lead to the creation of passive relays that are more efficient and capable of handling a wider range of frequencies and signal strengths.

3.Integration with IoT

The integration of passive relays with Internet of Things (IoT) networks could enable seamless communication between underwater sensors and surface systems, expanding monitoring and data collection capabilities.

4.Environmental Adaptability

Future research could focus on developing relays that can adapt to varying environmental conditions, such as changes in water salinity or temperature, to maintain reliable communication.



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5. Cost Reduction

Advances in technology and manufacturing processes might reduce the cost of passive relay systems, making them more accessible for a range of applications.

6.Scalable Solutions

As technology progresses, scalable solutions for deploying passive relays in large networks could be developed, enhancing the scope and effectiveness of air-to-underwater communication systems.

7.Interdisciplinary Innovations

Collaborations across fields like acoustics, materials science, and engineering could lead to new breakthroughs, improving the overall performance and functionality of photoacoustic communication systems.



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- In this paper, a photoacoustic air-to-underwater communication method based on low-cost passive relays is developed.
- Through numerical simulation and experimental verification, it is concluded that the proposed communication is capable of not only minimizing the energy loss both in water and in the air-water interface, but also relieving the difficulty of target seeking.
- The method is a promising way to communicate from the air to underwater, especially in complex water conditions.



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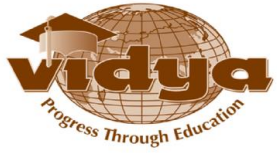
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- [1] Z. Ji, Y. Fu, J. Li, Z. Zhao and W. Mai, "Photoacoustic Communication from the Air to Underwater Based on Low-Cost Passive Relays," in IEEE Communications Magazine, vol. 59, no. 1, pp. 140-143, January 2021, doi: 10.1109/MCOM.001.2000607.
- [2] Mahmud M, Islam MS, Ahmed A, Younis M, Choa FS. Cross-Medium Photoacoustic Communications: Challenges, and State of the Art. Sensors (Basel). 2022 Jun 1;22(11)
- [3] Md Shafiqul Islam, Mohamed Younis, Muntasir Mahmud, Gary Carter, Fow-Sen Choa, A peak detection based OOK photoacoustic modulation scheme for air to underwater communication, Optics Communications, Volume 529, 2023, 129078, ISSN 0030-4018,
- [4] X. Wang et al., "Microwave-Induced Thermoacoustic Communications", IEEE Trans. Microwave Theory Tech., vol. 65, no.9, 2017, pp.3369-78



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